



Naziru HALILU

DIGITAL AGRICULTURE

PROFILE

Agricultural engineer with a long term research focus on the digitalization of agriculture, spanning computer vision, geospatial machine learning, agricultural control systems, and high throughput phenotyping, grounded in a strong foundation in agricultural machinery, mechanization, and farm power systems.

EMPLOYMENT HISTORY

EMJMD Research Scholar

Sep 2026 to Jan 2027

Agricultural University of Athens, Greece

- Term 3 focus: efficient water use, smart irrigation (30 ECTS). Courses: Modelling Crop Water Requirements, Smart Design and Management of Irrigation Systems, Modelling for Sustainable Irrigation, Agri-environmental Legislation and Policies, Technology Sharing and Transfer for Entrepreneurship, Practical Exercise Integrating Disciplines III.
- Continued graduate level research training in smart design and management of irrigation systems.
- Built NaCROP, a modular Python/Tkinter pipeline for evapotranspiration, soil water balance, irrigation scheduling, and a four step crop growth simulation for five smallholder crops around Zaria, Nigeria. [Code: GitHub](#)

Summer Intern, Computer Vision Group

Jun to Aug 2026

Dept. of Computer Engineering, ABU, Zaria

- Accomplished cross condition transfer of crop disease recognition models, as measured by 64.4 percent classification accuracy (macro AUC 0.882 against a 5.3 percent chance baseline) across 19 disease classes on 1,543 field images, by extending the Geometry Agnostic Contrastive Learning (GACL) framework with four new geometry invariant neural components. First author paper under review at Biosystems Engineering, Elsevier. [Code: GitHub](#)

EMJMD Research Scholar

Feb to Jun 2026

UTAD, Portugal

- Term 2 focus: precision agriculture for sustainable crop production (30 ECTS). Courses: Precision Agriculture, Identifying Soil and Crop Variability, Precision Agriculture Machinery, Preventive Management Strategy, Reactive Management Strategy, Practical Exercise Integrating Disciplines II.
- Accomplished cross system precision agriculture prediction and input cost reduction, as measured by R squared between 0.978 and 0.995 for grape ripening and maize yield regressions and a 2.69 tonne cut in lime demand. [Code: GitHub](#)
- Accomplished more reliable soil nutrient and yield mapping via Multitask Singularity Regression Kriging (MSRK), as measured by a roughly 20 percent reduction in residual kriging uncertainty across 17,864 samples. [Code: GitHub](#)
- Accomplished predictive harvest date forecasting for grape ripening, as measured by R squared of 0.9784, at a six hectare UTAD vineyard. [Code: GitHub](#)
- Accomplished a costed, spatially targeted pest management programme for grape moth across a 42.97 hectare Douro vineyard estate. [Code: GitHub](#)

EMJMD Research Scholar

Sep 2025 to Jan 2026

Public University of Navarre, Spain

- Term 1 focus: technological bases for agricultural sustainability (30 ECTS). Courses: Data Science and Analytics in Agriculture, Geospatial Data Analysis, Geoinformation and Remote Sensing, Sensors and the Internet of Things, Agricultural Systems Modelling, Practical Exercise Integrating Disciplines I.
- Accomplished real time, field deployable crop health screening, as measured by 33.8 percent balanced accuracy against a 4.55 percent chance baseline, by building a fourteen module RGB/multispectral vegetation indexing pipeline. [Code: GitHub](#)
- Accomplished radiometrically consistent multi temporal vegetation monitoring, as measured by a 39 and 31 percent reduction in temporal NDVI variance, by designing STADOS, a five stage adaptive dark object subtraction framework. [Code: GitHub](#)
- Accomplished accurate land suitability mapping for livestock slurry application, as measured by RMSE of 7.78 and R squared of 0.77 across 14.35 square kilometres near Tudela, Spain. [Code: GitHub](#)
- Accomplished a 47.8 percent reduction in annual soil loss via a Bayesian grounded digital twin for gully erosion biocontrol, hold out R squared of 0.90. [Code: GitHub](#)
- Built a six notebook vector/raster GIS and Sentinel 1/2 remote sensing course package in Python and Google Earth Engine. [Code: GitHub](#)

Undergraduate Project Assistantships

Mar 2024 to Mar 2025

ABU, Nigeria

- Contributed data collection, statistical analysis, and literature review across concurrent research projects in water and irrigation systems, soil and erosion assessment, and postharvest and animal systems.

Project Assistant, Remote Data Lead

2024 to 2025

CONTACT

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[Zenodo record list](#)

halilunaziru73-creator.github.io

DETAILS

Location: Pamplona, Spain

Nationality: Nigerian

EDUCATION

2025 to expected Jun/Aug 2027

EMJMD, Sustainable Agriculture (SUSTAGRI)

UPNA, UTAD, AUA

2019 to 2024

B.Eng, Agric. and Bioresources Engineering

ABU, Nigeria. First Class Honours, CGPA 4.77 of 5.00, Top 1 percent

2016 to 2018

National Diploma, Agric. Engineering Technology

ABU. Distinction, CGPA 3.78 of 4.00, Top 1 percent

SKILLS

Python / Deep Learning



GIS and Remote Sensing



Computer Vision



Geostatistics



Agric. Machinery Systems



MATLAB / R



LANGUAGES

English (fluent)

MEMBERSHIPS

ASABE, NIAE, NSE, Pupil Engineer
COREN

Naziru Halilu

Digital Agriculture Researcher

RESEARCH IMPACT

- 14 Open repositories
- 14 Zenodo DOIs archived
- 4 Peer reviewed papers
- 6 Under review
- 10 Awards and honors

ORGANIZATIONS

- Agricultural University of Athens
- UTAD, Portugal
- Public University of Navarre
- Ahmadu Bello University (ABU)
- IAR-ABU
- NAERLS
- Yankari Nasarawa Farms
- BOMO Basin Project
- NCAM, Ilorin
- Takai Academy

CERTIFICATES

- EMJMD certificate / enrolment
- B.Eng transcript / certificate
- ND transcript / certificate

IAR-ABU, Nigeria

- Engineered adaptive irrigation scheduling algorithms fusing real time soil moisture and microclimate telemetry for automated greenhouse control.

Student Research Assistant, Precision Crop Systems

2022 to 2024

NAERLS, Nigeria

- Trained and validated CNN crop stress detection models reaching 78 percent accuracy on multispectral UAV imagery.

Project Assistant, Smart Irrigation and Water Systems

2021 to 2022

BOMO Basin Project, Nigeria

- Implemented closed loop control logic for a one hectare smart drip irrigation array.

Research Intern

2017 to 2020

NCAM, Ilorin and IAR, Zaria, Nigeria

- Built Python data logging pipelines to benchmark mechanization efficiency; supported baseline hydrological research.

TEACHING, LEADERSHIP AND COMMUNITY ENGAGEMENT

Undergraduate Course Tutor

2020 to 2024

ABU Zaria (NAMET)

- Supplementary tutorials in Differential Equations and Transforms, Irrigation System Design, and Computer Applications in Agric. Engineering.

Secretary General, NAAESB

2023 to 2024

National Assoc. of Agric. Engineering Students

- Directed operational targets and inter chapter scientific meetups for the national student body.

Director, Environmental Protection Agency

2023 to 2024

ABU Students' Union Government

- Directed campus environmental protection and sanitation initiatives.

Deputy Coordinator

2020 to 2025

KOJYAF, Zango Shanu, Zaria

- Coordinated tutorial and mentorship programs preparing secondary students for JAMB entrance examinations.

Physics and Mathematics Teacher

2022 to 2023

Takai Academy, Zaria, Nigeria

- Instructed senior classes in classical physics and computational principles.

PUBLICATIONS

- Halilu, N., and Sulaiman, J. (2026). Real Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. IJRSI, 13(7), 2951 to 2989. DOI
- Halilu, N. (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. IJRSI, 13(7), 2920 to 2950. DOI
- Halilu, N., and Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. IJRSI, 13(7), 2990 to 3020. DOI
- Ismail, K., and Halilu, N. (2024). AI Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. Proc. AfCPA, 2(1), 100 to 120.
- Halilu, N., and Sulaiman, J. Geometry Agnostic Contrastive Learning (GACL). Biosystems Engineering, Elsevier. Under review. DOI
- Halilu, N. (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (Ipomoea spp.). Environmental Monitoring and Assessment, Springer Nature. Under review. DOI
- Halilu, N., and Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. European Journal of Agronomy, Elsevier. Under review. DOI
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time Series Modelling and Control Systems in Precision Irrigation. Zenodo Preprint. DOI
- Halilu, N. (2026). Multitask Singularity Regression Kriging for Spatio Temporal Soil NPK Crop Stress Yield Prediction. Zenodo Preprint. DOI

AWARDS AND HONORS

Erasmus Mundus Joint Master Scholarship, 2025 · Best Graduating Student, B.Eng, 2024 · Nigerian Federal Government Award for Academic Excellence, 2024 · Best Undergraduate Research Paper Award, ABU, 2024 · Best Overall Graduating Student, National Diploma, 2019 · PTDF Undergraduate Fellow, 2023.